

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2023-24

Paper: BVCA-304 — Java Applet Programming

Time: Three Hours

Maximum Marks: 70

*Instructions: Attempt all questions. Symbols have their usual meanings.***SECTION — A**

Long Answer type questions to be answered in about 500 words each. Attempt any two: **12 × 2 = 24 Marks**

Question 1. What do you understand by the concept of Object Oriented Language? Explain how Java is specified as an Object Oriented Language.

Question 2. Explain about **for**, **while** and **do-while** loop structures in Java with examples. Why are they categorized as entry-controlled and exit-controlled loops?

Question 3. What are operators? Explain all the Operators supported by the Java programming language.

Question 4. Discuss the Switch Statement. Write a program to calculate the percentage of total marks obtained in 5 subjects by a student and display the equivalent grade as S, A, B, C, D, or F. (Grade criteria: 91–100% as S grade; 81–90% as A grade; 71–80% as B grade and so on. F grade represents Fail).

SECTION — B

Question 5. Attempt any six parts (Answer in about 200 words each): **6 × 6 = 36 Marks**

- (a) What do you understand by classes and objects? How are objects created from a class?
- (b) What are arrays? How will you implement one-dimensional and two-dimensional arrays?
- (c) Define JDK and JVM. Why is Java considered a platform-independent language?
- (d) Differentiate between static, instance, and local variables.
- (e) Discuss break and continue statements in Java.
- (f) Write a program to count the digits of a number input by the user.
- (g) Write a Java program to find the area and circumference of a circle using the concept of objects and classes.
- (h) Discuss any four inbuilt methods for String manipulation given in the Java library.

SECTION — C

Question 6. Attempt all parts:**10 × 1 = 10 Marks**

- (a) What does an Eclipse IDE allow?
 - (a) Editing the program
 - (b) Compiling the program
 - (c) Both Editing and Compiling the program
 - (d) Editing, Building, Debugging and Compiling the program
- (b) Which keyword is used for accessing the features of a package?
 - (a) package
 - (b) import
 - (c) extends
 - (d) export
- (c) Which of these values is returned by the `read()` method if end of file (EOF) is encountered?
 - (a) 0
 - (b) 1
 - (c) -1
 - (d) Null
- (d) Which one of the following is not a Java feature?
 - (a) Object-oriented
 - (b) Use of pointers
 - (c) Portable
 - (d) Dynamic and Extensible
- (e) JRE stands for _____.
 - (a) Java Run Ecosystem
 - (b) JDK Runtime Environment
 - (c) Java Runtime Environment
 - (d) None of these
- (f) Which class in Java is used to take input from the user?
 - (a) Scanner
 - (b) Input

- (c) Applier
 - (d) None of these
- (g) Automatic type conversion is possible in which of the following cases?
- (a) Byte to int
 - (b) Int to long
 - (c) Long to int
 - (d) Short to int
- (h) Identify the output of the following program:

```
String str = "abcde";  
System.out.println(str.substring(1, 3));
```

- (a) abc
 - (b) bc
 - (c) bcd
 - (d) cd
- (i) Find the output of the following code:

```
int ++a = 100;  
System.out.println(++a);
```

- (a) 101
 - (b) Compile error as ++a is not a valid identifier
 - (c) 100
 - (d) None of these
- (j) Find the output of the following code:

```
public class Solution {  
    public static void main(String args[]) {  
        int i;  
        for (i = 1; i < 6; i++) {  
            if (i > 3) continue;  
        }  
        System.out.println(i);  
    }  
}
```

- (a) 3
- (b) 4
- (c) 5
- (d) 6

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